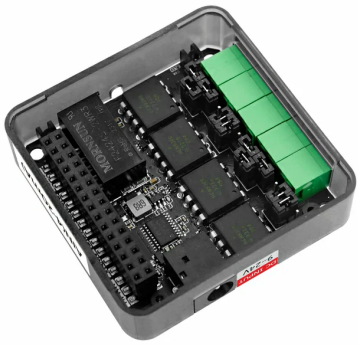
Module13.2 AIN4-20mA

SKU:M133

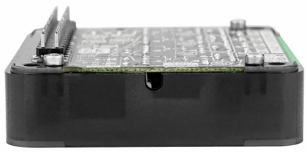


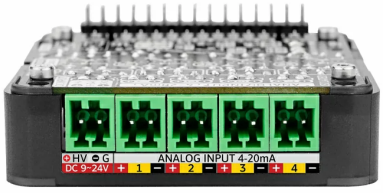


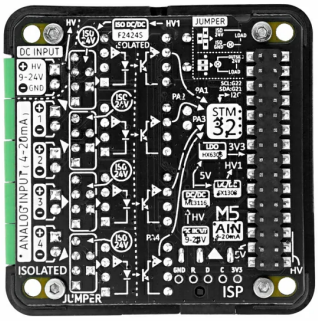
















Description

Module13.2 AIN4-20mA is a **four-channel 4 ~ 20 mA** current-type analog acquisition module. It uses the **STM32G030F6** main control chip and a dedicated isolated acquisition chip, communicating with the M5 host via **I2C**. It supports switching between internal or external power supply through jumper caps. The onboard power isolation chip and built-in operational amplifier circuit ensure accurate measurement of external current sensors and the accuracy of the signal and system safety. The **DC-JACK** interface and corresponding **DC-DC** boost circuit provide power to the entire device. **It is suitable for power system equipment monitoring, motor control, energy management, automation, and industrial process control.**

Features

- STM32G030F6® 32-bit Cortex®-M0+ CPU
- I2C communication
- Supports four 2-wire or 4-wire sensors, switchable via jumper caps
- Built-in electrical isolation chip
- Supports Arduino, UIFlow, and other programming platforms

Includes

- 1 × AIN4-20mA Module 13.2
- 12 × Jumper caps
- 5 × KF2EDGK-2.54-2P connection terminals

Applications

- Power system equipment monitoring
- Motor control
- Energy management
- Automation and industrial process control

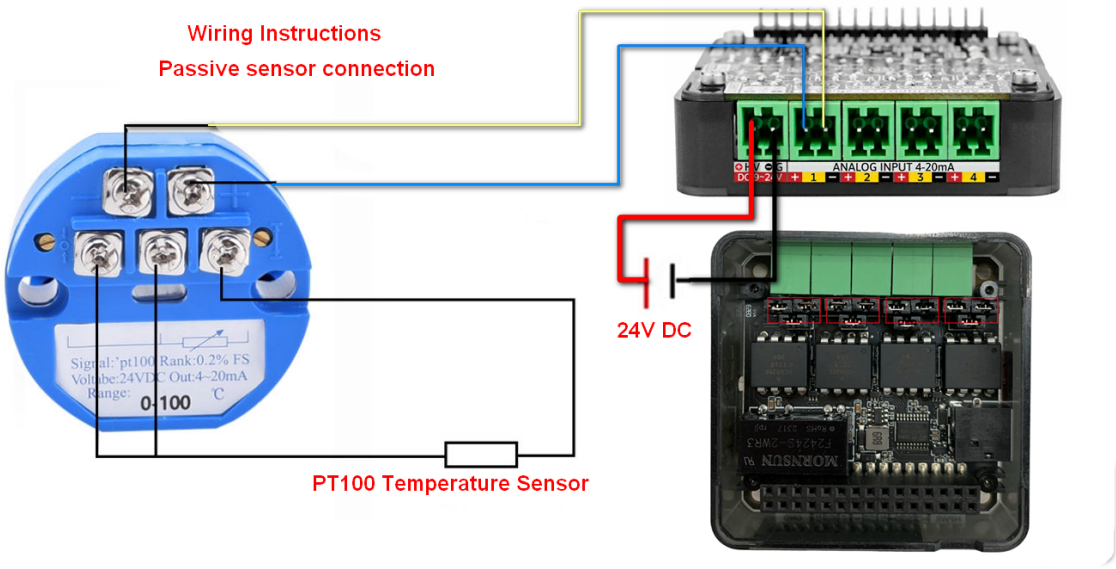
Specifications

Specification	Parameter
MCU	STM32G030F6P6
Signal Isolation Chip	HCNR200
Power Isolation Chip	F2424S-2WR3
Operational Amplifier	SGM321YC5/TR
Communication Interface	I2C Communication @ 0x55
IN+ and IN- Input Impedance	200Ω
Channel Interface Specification	KF2EDGR-2.54-2P
Operating Temperature	0 ~ 40°C
External DC Power Supply	DC 9 ~ 24V
Product Size	54.0 x 54.0 x 19.7mm
Product Weight	26.5g
Package Size	95.0 x 66.0 x 26.0mm
Gross Weight	38.4g

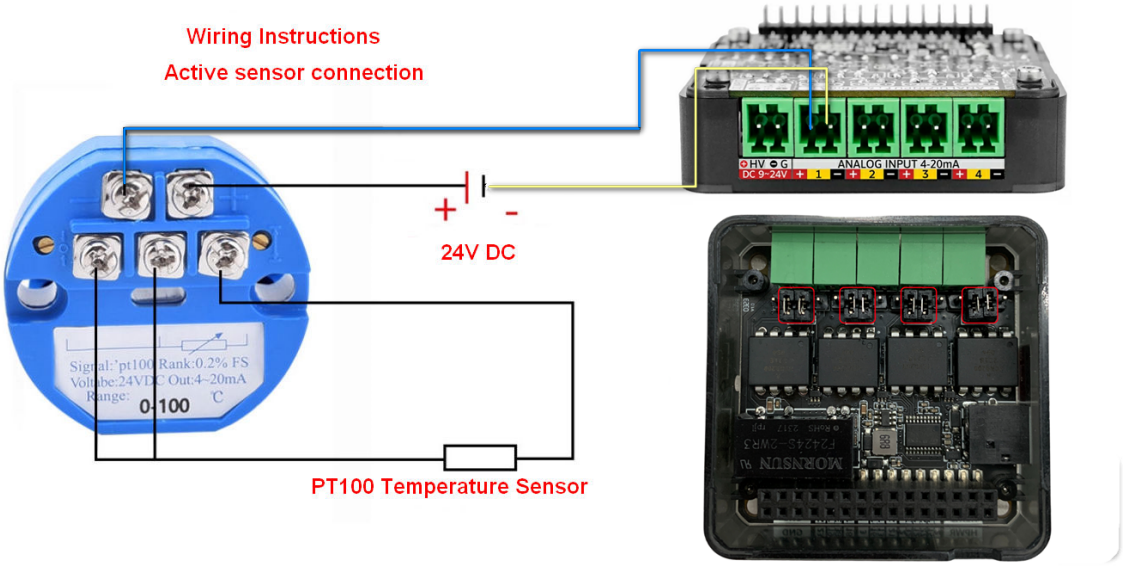
Learn

Jumper Cap Connection and Instructions

- When using a **passive current-type sensor**, please connect the DC 24V power supply input, connect the sensor signal to **IN+ and IN-**, and adjust the jumper cap as shown below:

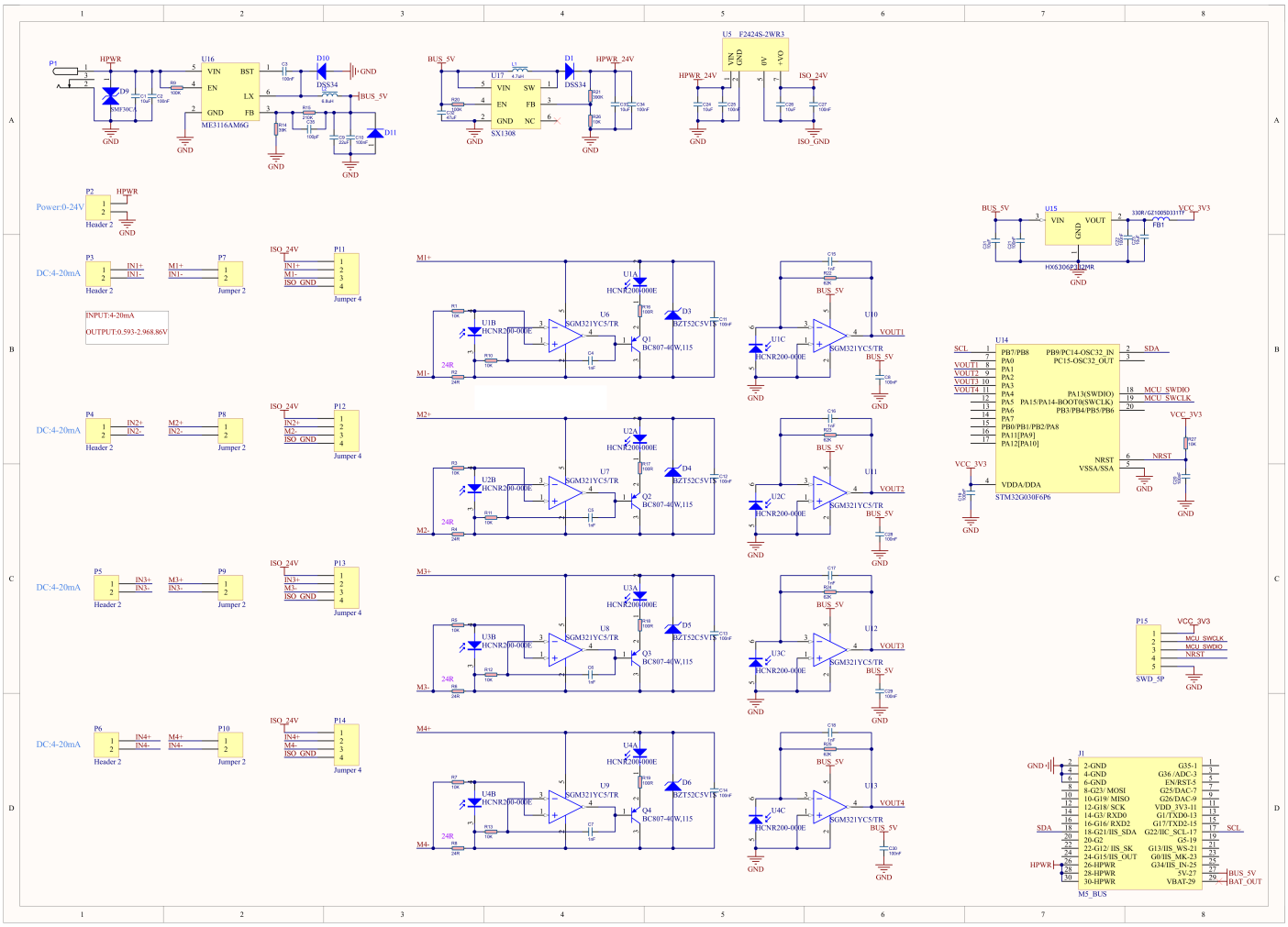


- When using an **active current-type sensor**, connect the sensor signal to **IN+ and IN-**, and adjust the jumper cap as shown below:



Schematics

- Module13.2 AIN4-20mA Schematics PDF



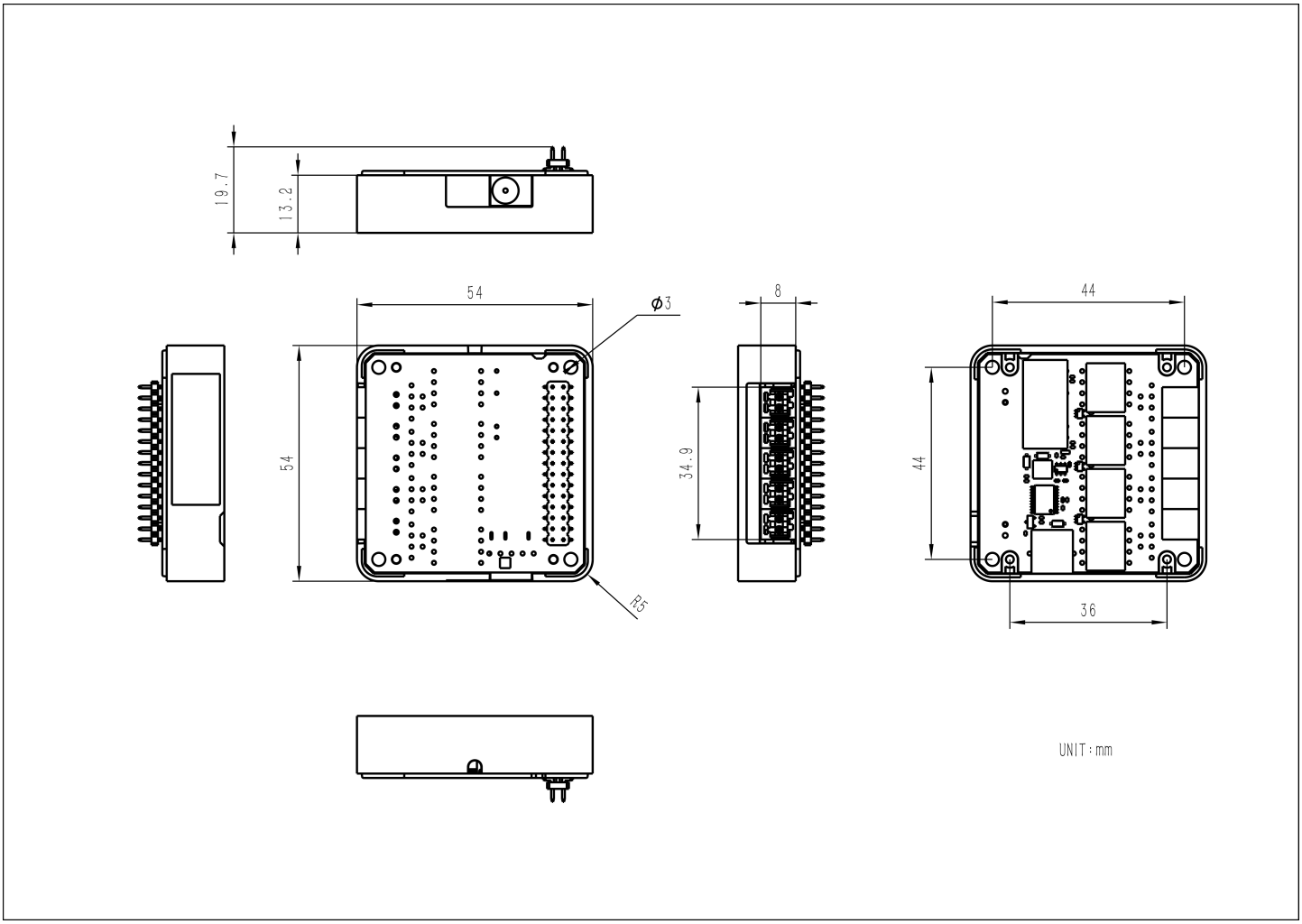
PinMap

M5-Bus

PIN	LEFT	RIGHT	PIN
GND	1	2	
GND	3	4	
GND	5	6	
	7	8	
	9	10	
	11	12	
	13	14	
	15	16	
SDA	17	18	SCL
	19	20	
	21	22	
	23	24	
HPWR	25	26	
HPWR	27	28	BUS_5V
HPWR	29	30	BAT_OUT

Model Size

- Module13.2 AIN4-20mA Model Size PDF



Datasheets

- STM32G030F6 Datasheet
- HCNR200 Datasheet
- SGM321YC5 Datasheet

Softwares

Arduino

- Module13.2 AIN4-20mA Arduino Library
- Module13.2 AIN4-20mA Arduino Tutorial

UiFlow1

- Module13.2 AIN4-20mA UiFlow1 Docs

UiFlow2

- Module13.2 AIN4-20mA UiFlow2 Docs

Internal Firmware

- Module13.2 AIN4-20mA Internal Firmware

Protocol

M5Stack Module 4-20mA I2C Protocol																	V1 (FW Version) 2023/1/30
REG MAP (Addr:0x55)		0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	note
Raw ADC (12Bits)	0x00 R	ADC1-L	ADC1-H	ADC2-L	ADC2-H	ADC3-L	ADC3-H	ADC4-L	ADC4-H									
Current	0x20 R	Current 1-L	Current 1-H	Current 2-L	Current 2-H	Current 3-L	Current 3-H	Current 4-L	Current 4-H									
CAL[1]	0x30 W	Ref Current -L	Ref Current -H	Ref Current -L	Ref Current -H	Ref Current -L	Ref Current -H	Ref Current -L	Ref Current -H									
Firmware Version	0xF0 R															Version		**Version:** firmware version number
I2C Address	0xF0 R																Address	**Address:** I2C Address

[1] Calibration method:
(1) Generate a standard current using a 4-20mA signal source, such as 10.00mA.
(2) Write the current value into the register, but note that this value needs to be multiplied by 100. For example, for 10.00mA, the calibration value should be 1000. Therefore, Ref Current-L should be set to 0xE8, and Ref Current-H should be set to 0x03.

Video